# 1

# Introduction

## 1.1 Historical Perspective

This book gives a comprehensive treatment of optimal $\mathcal{H}_2$ and $\mathcal{H}_\infty$ control theory and an introduction to the more general subject of robust control. Since the central subject of this book is *state-space* $\mathcal{H}_\infty$ optimal control, in contrast to the approach adopted in the famous book by Francis [1987]: *A Course in* $\mathcal{H}_\infty$ *Control Theory*, it may be helpful to provide some historical perspective of the state-space $\mathcal{H}_\infty$ control theory to be presented in this book. This section is not intended as a review of the literature in $\mathcal{H}_\infty$ theory or robust control, but rather only an attempt to outline some of the work that most closely touches on our approach to state-space $\mathcal{H}_\infty$. Hopefully our lack of written historical material will be somewhat made up for by the pictorial history of control shown in Figure 1.1. Here we see how the practical but classical methods yielded to the more sophisticated modern theory. Robust control sought to blend the best of both worlds. The strange creature that resulted is the main topic of this book.

The $\mathcal{H}_\infty$ optimal control theory was originally formulated by Zames [1981] in an input-output setting. Most solution techniques available at that time involved analytic functions (Nevanlinna-Pick interpolation) or operator-theoretic methods [Sarason, 1967; Adamjan *et al.*, 1978; Ball and Helton, 1983]. Indeed, $\mathcal{H}_\infty$ theory seemed to many to signal the beginning of the end for the state-space methods which had dominated control for the previous 20 years. Unfortunately, the standard frequency-domain approaches to $\mathcal{H}_\infty$ started running into significant obstacles in dealing with multi-input multi-output (MIMO) systems, both mathematically and computationally, much as the $\mathcal{H}_2$ (or LQG) theory of the 1950's had.


Figure 1.1: A picture history of control

Not surprisingly, the first solution to a general rational MIMO $\mathcal{H}_\infty$ optimal control problem, presented in Doyle [1984], relied heavily on state-space methods, although more as a computational tool than in any essential way. The steps in this solution were as follows: parameterize all internally-stabilizing controllers via [Youla *et al.*, 1976]; obtain realizations of the closed-loop transfer matrix; convert the resulting model-matching problem into the equivalent $2 \times 2$-block general distance or best approximation problem involving mixed Hankel-Toeplitz operators; reduce to the Nehari problem (Hankel only); solve the Nehari problem by the procedure of Glover [1984]. Both [Francis, 1987] and [Francis and Doyle, 1987] give expositions of this approach, which will be referred to as the "1984" approach.

In a mathematical sense, the 1984 procedure "solved" the general rational $\mathcal{H}_\infty$ optimal control problem and much of the subsequent work in $\mathcal{H}_\infty$ control theory focused on the $2 \times 2$-block problems, either in the model-matching or general distance forms. Unfortunately, the associated complexity of computation was substantial, involving several Riccati equations of increasing dimension, and formulae for the resulting controllers tended to be very complicated and have high state dimension. Encouragement came from Limebeer and Hung [1987] and Limebeer and Halikias [1988] who showed, for problems transformable to $2 \times 1$-block problems, that a subsequent minimal realization of the controller has state dimension no greater than that of the generalized plant $G$. This suggested the likely existence of similarly low dimension optimal controllers in the general $2 \times 2$ case.

Additional progress on the $2 \times 2$-block problems came from Ball and Cohen [1987], who gave a state-space solution involving 3 Riccati equations. Jonckheere and Juang [1987] showed a connection between the $2 \times 1$-block problem and previous work by Jonckheere and Silverman [1978] on linear-quadratic control. Foias and Tannenbaum [1988] developed an interesting class of operators called skew Toeplitz to study the $2 \times 2$-block problem. Other approaches have been derived by Hung [1989] using an interpolation theory approach, Kwakernaak [1986] using a polynomial approach, and Kimura [1988] using a method based on conjugation.

The simple state space $\mathcal{H}_\infty$ controller formulae to be presented in this book were first derived in Glover and Doyle [1988] with the 1984 approach, but using a new $2 \times 2$-block solution, together with a cumbersome back substitution. The very simplicity of the new formulae and their similarity with the $\mathcal{H}_2$ ones suggested a more direct approach.

Independent encouragement for a simpler approach to the $\mathcal{H}_\infty$ problem came from papers by Petersen [1987], Khargonekar, Petersen, and Zhou [1990], Zhou and Khargonekar [1988], and Khargonekar, Petersen, and Rotea [1988]. They showed that for the state-feedback $\mathcal{H}_\infty$ problem one can choose a constant gain as a (sub)optimal controller. In addition, a formula for the state-feedback gain matrix was given in terms of an algebraic Riccati equation. Also, these papers established connections between $\mathcal{H}_\infty$-optimal control, quadratic stabilization, and linear-quadratic differential games.

The landmark breakthrough came in the DGKF paper (Doyle, Glover, Khargonekar, and Francis [1989]). In addition to providing controller formulae that are simple and expressed in terms of plant data as in Glover and Doyle [1988], the methods in that paper are a fundamental departure from the 1984 approach. In particular, the Youla parameterization and the resulting $2 \times 2$-block model-matching problem of the 1984 solution are avoided entirely; replaced by a more purely state-space approach involving observer-based compensators, a pair of $2 \times 1$ block problems, and a separation argument. The operator theory still plays a central role (as does Redheffer's work [Redheffer, 1960] on linear fractional transformations), but its use is more straightforward. The key to this was a return to simple and familiar state-space tools, in the style of Willems [1971], such as completing the square, and the connection between frequency domain inequalities (e.g $\|G\|_\infty < 1$), Riccati equations, and spectral factorizations. This book in some sense can be regarded as an expansion of the DGKF paper.

The state-space theory of $\mathcal{H}_\infty$ can be carried much further, by generalizing time-invariant to time-varying, infinite horizon to finite horizon, and finite dimensional to infinite dimensional. A flourish of activity has begun on these problems since the publication of the DGKF paper and numerous results have been published in the literature, not surprising, many results in DGKF paper generalize *mutatis mutandis*, to these cases, which are beyond the scope of this book.

## 1.2 How to Use This Book

This book is intended to be used either as a graduate textbook or as a reference for control engineers. With the second objective in mind, we have tried to balance the broadness and the depth of the material covered in the book. In particular, some chapters have been written sufficiently self-contained so that one may jump to those special topics without going through all the preceding chapters, for example, Chapter 13 on algebraic Riccati equations. Some other topics may only require some basic linear system theory, for instance, many readers may find that it is not difficult to go directly to Chapters $9 \sim 11$. In some cases, we have tried to collect some most frequently used formulas and results in one place for the convenience of reference although they may not have any direct connection with the main results presented in the book. For example, readers may find that those matrix formulas collected in Chapter 2 on linear algebra convenient in their research. On the other hand, if the book is used as a textbook, it may be advisable to skip those topics like Chapter 2 on the regular lectures and leave them for students to read. It is obvious that only some selected topics in this book can be covered in an one or two semester course. The specific choice of the topics depends on the time allotted for the course and the preference of the instructor. The diagram in Figure 1.2 shows roughly the relations among the chapters and should give the users some idea for the selection of the topics. For example, the diagram shows that the only prerequisite for Chapters 7 and 8 is Section 3.9 of Chapter 3 and, therefore, these two chapters alone may be used as a short course on model reductions. Similarly, one only needs the knowledge of Sections 13.2 and 13.6 of Chapter 13 to understand Chapter 14. Hence one may only cover those related sections of Chapter 13 if time is the factor. The book is separated roughly into the following subgroups:

- Basic Linear System Theory: Chapters $2 \sim 3$.
- Stability and Performance: Chapters $4 \sim 6$.
- Model Reduction: Chapters $7 \sim 8$.
- Robustness: Chapters $9 \sim 11$.
- $\mathcal{H}_2$ and $\mathcal{H}_\infty$ Control: Chapters $12 \sim 19$.
- Lagrange Method: Chapter 20.
- Discrete Time Systems: Chapter 21.

In view of the above classification, one possible choice for an one-semester course on robust control would cover Chapters $4 \sim 5, 9 \sim 11$ or $4 \sim 11$ and an one-semester advanced course on $\mathcal{H}_2$ and $\mathcal{H}_\infty$ control would cover (parts of) Chapters $12 \sim 19$. Another possible choice for an one-semester course on $\mathcal{H}_\infty$ control may include Chapter 4, parts of Chapter 5 ($5.1 \sim 5.3, 5.5, 5.7$), Chapter 10, Chapter 12 (except Section 12.6), parts of Chapter 13 ($13.2, 13.4, 13.6$), Chapter 15 and Chapter 16. Although Chapters $7 \sim 8$ are very much independent of other topics and can, in principle, be studied at any

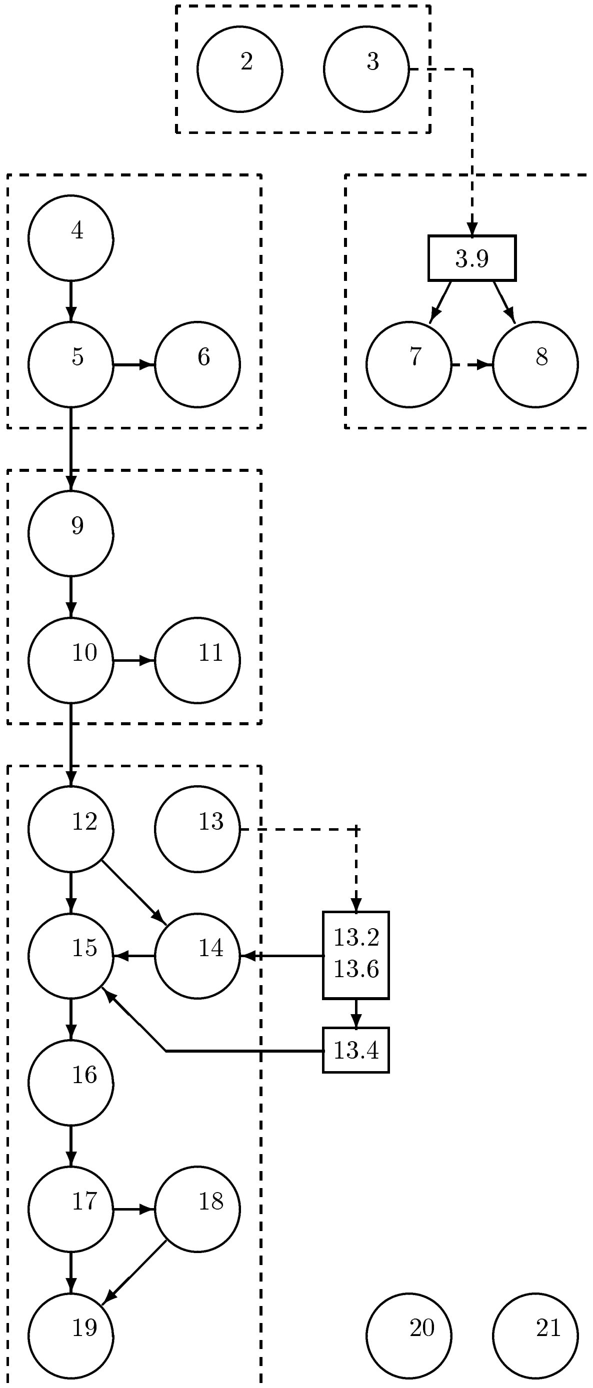



Figure 1.2: Relations among the chapters

stage with the background of Section 3.9, they may serve as an introduction to sources of model uncertainties and hence to robustness problems.

| Robust Control | $\mathcal{H}_\infty$ Control | Advanced $\mathcal{H}_\infty$ Control | Model & Controller Reductions |
|---|---|---|---|
| 4 | 4 | 12 | 3.9 |
| 5 | 5.1∼5.3,5.5,5.7 | 13.2,13.4,13.6 | 7 |
| 6* | 10 | 14 | 8 |
| 7* | 12 | 15 | 5.4,5.7 |
| 8* | 13.2,13.4,13.6 | 16 | 10.1 |
| 9 | 15 | 17* | 16.1,16.2 |
| 10 | 16 | 18* | 17.1 |
| 11 | | 19* | 19 |

Table 1.1: Possible choices for an one-semester course (* chapters may be omitted)

Table 1.1 lists several possible choices of topics for an one-semester course. A course on model and controller reductions may only include the concept of $\mathcal{H}_\infty$ control and the $\mathcal{H}_\infty$ controller formulas with the detailed proofs omitted as suggested in the above table.

## 1.3 Highlights of The Book

The key results in each chapter are highlighted below. Note that some of the statements in this section are not precise, they are true under certain assumptions that are not explicitly stated. Readers should consult the corresponding chapters for the exact statements and conditions.

Chapter 2 reviews some basic linear algebra facts and treats a special class of matrix dilation problems. In particular, we show

$$\min_X \left\| \begin{bmatrix} X & B \\ C & A \end{bmatrix} \right\| = \max \left\{ \left\| \begin{bmatrix} C & A \end{bmatrix} \right\|, \left\| \begin{bmatrix} B \\ A \end{bmatrix} \right\| \right\}$$

and characterize all optimal (suboptimal) $X$.

Chapter 3 reviews some system theoretical concepts: controllability, observability, stabilizability, detectability, pole placement, observer theory, system poles and zeros, and state space realizations. Particularly, the balanced state space realizations are studied in some detail. We show that for a given stable transfer function $G(s)$ there is a state space realization $G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ such that the controllability Gramian $P$ and the observability Gramian $Q$ defined below are equal and diagonal: $P = Q = \Sigma = \operatorname{diag}(\sigma_1, \sigma_2, \ldots, \sigma_n)$ where

$$AP + PA^* + BB^* = 0$$

$$A^*Q + QA + C^*C = 0.$$

Chapter 4 defines several norms for signals and introduces the $\mathcal{H}_2$ spaces and the $\mathcal{H}_\infty$ spaces. The input/output gains of a stable linear system under various input signals are characterized. We show that $\mathcal{H}_2$ and $\mathcal{H}_\infty$ norms come out naturally as measures of the worst possible performance for many classes of input signals. For example, let

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & 0 \end{array}\right] \in \mathcal{RH}_\infty, \quad g(t) = Ce^{At}B$$

Then $\|G\|_\infty = \sup \dfrac{\|g * u\|_2}{\|u\|_2}$ and $\sigma_1 \le \|G\|_\infty \le \int_0^\infty \|g(t)\|\, dt \le 2\sum_{i=1}^{n} \sigma_i$. Some state space methods of computing real rational $\mathcal{H}_2$ and $\mathcal{H}_\infty$ transfer matrix norms are also presented:

$$\|G\|_2^2 = \operatorname{trace}(B^*QB) = \operatorname{trace}(CPC^*)$$

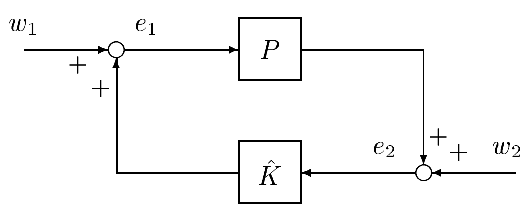
and

$$\|G\|_\infty = \max\{\gamma : \ H \text{ has an eigenvalue on the imaginary axis}\}$$

where

$$H = \begin{bmatrix} A & BB^*/\gamma^2 \\ -C^*C & -A^* \end{bmatrix}.$$

Chapter 5 introduces the feedback structure and discusses its stability and performance properties.




We show that the above closed-loop system is internally stable if and only if

$$\begin{bmatrix} I & -\hat{K} \\ -P & I \end{bmatrix}^{-1} = \begin{bmatrix} I + \hat{K}(I - P\hat{K})^{-1}P & \hat{K}(I - P\hat{K})^{-1} \\ (I - P\hat{K})^{-1}P & (I - P\hat{K})^{-1} \end{bmatrix} \in \mathcal{RH}_\infty.$$

Alternative characterizations of internal stability using coprime factorizations are also presented.

Chapter 6 introduces some multivariable versions of the Bode's sensitivity integral relations and Poisson integral formula. The sensitivity integral relations are used to study the design limitations imposed by bandwidth constraint and the open-loop unstable poles, while the Poisson integral formula is used to study the design constraints imposed by the non-minimum phase zeros. For example, let $S(s)$ be a sensitivity function, and let $p_i$ be the right half plane poles of the open-loop system and $\eta_i$ be the corresponding pole directions. Then we show that

$$\int_0^\infty \ln \overline{\sigma}(S(j\omega))d\omega = \pi\lambda_{\max}\left(\sum_i (\mathrm{Re}p_i)\eta_i\eta_i^*\right) + H_1, \quad H_1 \geq 0.$$

This equality shows that the design limitations in multivariable systems are dependent on the directionality properties of the sensitivity function as well as those of the poles (and zeros), in addition to the dependence upon pole (and zero) locations which is known in single-input single-output systems.

Chapter 7 considers the problem of reducing the order of a linear multivariable dynamical system using the balanced truncation method. Suppose

$$G(s) = \left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ \hline C_1 & C_2 & D \end{array}\right] \in \mathcal{RH}_\infty$$

is a balanced realization with controllability and observability Gramians $P = Q = \Sigma = \mathrm{diag}(\Sigma_1, \Sigma_2)$

$$\begin{aligned} \Sigma_1 &= \mathrm{diag}(\sigma_1 I_{s_1}, \sigma_2 I_{s_2}, \ldots, \sigma_r I_{s_r}) \\ \Sigma_2 &= \mathrm{diag}(\sigma_{r+1} I_{s_{r+1}}, \sigma_{r+2} I_{s_{r+2}}, \ldots, \sigma_N I_{s_N}). \end{aligned}$$

Then the truncated system $G_r(s) = \left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & D \end{array}\right]$ is stable and satisfies an additive error bound:

$$\|G(s) - G_r(s)\|_\infty \leq 2\sum_{i=r+1}^N \sigma_i.$$

On the other hand, if $G^{-1} \in \mathcal{RH}_\infty$, and $P$ and $Q$ satisfy

$$PA^* + AP + BB^* = 0$$

$$Q(A - BD^{-1}C) + (A - BD^{-1}C)^*Q + C^*(D^{-1})^*D^{-1}C = 0$$

such that $P = Q = \mathrm{diag}(\Sigma_1, \Sigma_2)$ with $G$ partitioned compatibly as before, then

$$G_r(s) = \left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & D \end{array}\right]$$

is stable and minimum phase, and satisfies respectively the following relative and multiplicative error bounds:

$$\left\|G^{-1}(G - G_r)\right\|_\infty \leq \prod_{i=r+1}^N \left(1 + 2\sigma_i(\sqrt{1+\sigma_i^2} + \sigma_i)\right) - 1$$

$$\left\|G_r^{-1}(G - G_r)\right\|_\infty \leq \prod_{i=r+1}^N \left(1 + 2\sigma_i(\sqrt{1+\sigma_i^2} + \sigma_i)\right) - 1.$$

Chapter 8 deals with the optimal Hankel norm approximation and its applications in $\mathcal{L}_\infty$ norm model reduction. We show that for a given $G(s)$ of McMillan degree $n$ there is a $\hat{G}(s)$ of McMillan degree $r < n$ such that

$$\left\|G(s) - \hat{G}(s)\right\|_H = \inf \left\|G(s) - \hat{G}(s)\right\|_H = \sigma_{r+1}.$$

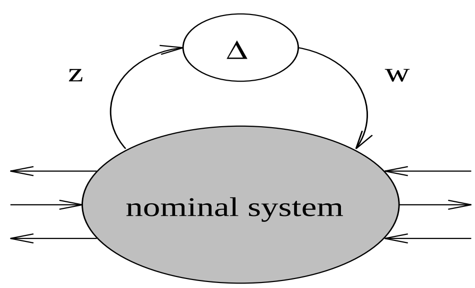
Moreover, there is a constant matrix $D_0$ such that

$$\left\|G(s) - \hat{G}(s) - D_0\right\|_\infty \leq \sum_{i=r+1}^{N} \sigma_i.$$

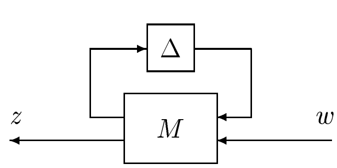
The well-known Nehari's theorem is also shown:

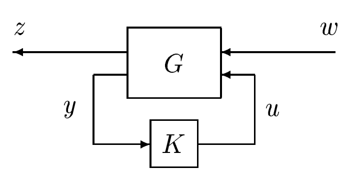
$$\inf_{Q \in \mathcal{RH}_\infty^-} \|G - Q\|_\infty = \|G\|_H = \sigma_1.$$

Chapter 9 derives robust stability tests for systems under various modeling assumptions through the use of a small gain theorem. In particular, we show that an uncertain system described below with an unstructured uncertainty $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty < 1$ is robustly stable if and only if the transfer function from $w$ to $z$ has $\mathcal{H}_\infty$ norm no greater than 1.

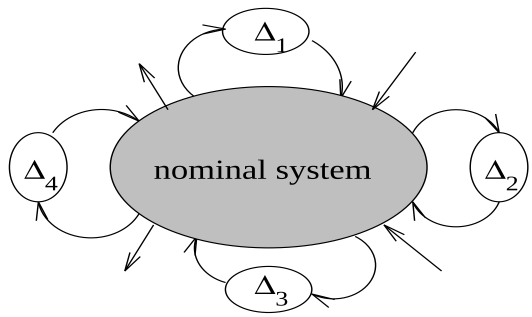



Chapter 10 introduces the linear fractional transformation (LFT). We show that many control problems can be formulated and treated in the LFT framework. In particular, we show that every analysis problem can be put in an LFT form with some structured $\Delta(s)$ and some interconnection matrix $M(s)$ and every synthesis problem can be put in an LFT form with a generalized plant $G(s)$ and a controller $K(s)$ to be designed.



Chapter 11 considers robust stability and performance for systems with multiple sources of uncertainties. We show that an uncertain system is robustly stable for all $\Delta_i \in \mathcal{RH}_\infty$ with $\|\Delta_i\|_\infty < 1$ if and only if the structured singular value ($\mu$) of the corresponding interconnection model is no greater than 1.



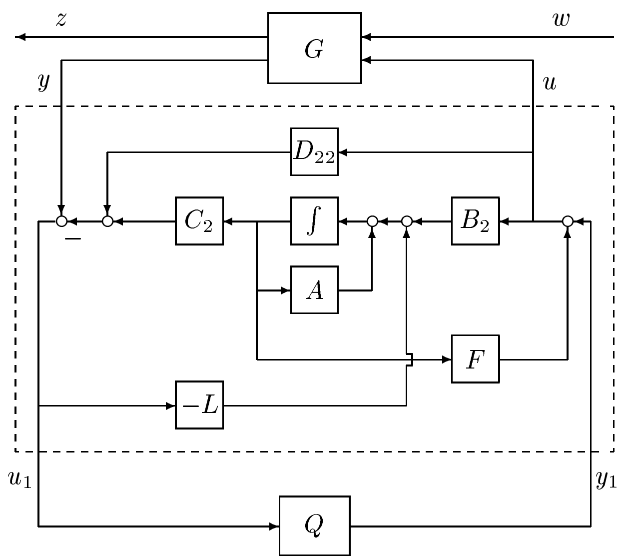

Chapter 12 characterizes all controllers that stabilize a given dynamical system $G(s)$ using the state space approach. The construction of the controller parameterization is done via separation theory and a sequence of special problems, which are so-called *full information (FI)* problems, *disturbance feedforward (DF)* problems, *full control* (FC) problems and *output estimation (OE)*. The relations among these special problems are established.

$$\begin{tikzcd}
\text{FI} \arrow[r, leftrightarrow, "dual"] \arrow[d, leftrightarrow, "\text{equivalent}"'] & \text{FC} \arrow[d, leftrightarrow, "\text{equivalent}"] \\
\text{DF} \arrow[r, leftrightarrow, "dual"] & \text{OE}
\end{tikzcd}$$

For a given generalized plant

$$G(s) = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array}\right]$$

we show that all stabilizing controllers can be parameterized as the transfer matrix from $y$ to $u$ below where $F$ and $L$ are such that $A + LC_2$ and $A + B_2F$ are stable.




Chapter 13 studies the *Algebraic Riccati Equation* and the related problems: the properties of its solutions, the methods to obtain the solutions, and some applications. In particular, we study in detail the so-called stabilizing solution and its applications in matrix factorizations. A solution to the following ARE

$$A^*X + XA + XRX + Q = 0$$

is said to be a stabilizing solution if $A + RX$ is stable. Now let

$$H := \begin{bmatrix} A & R \\ -Q & -A^* \end{bmatrix}$$

and let $\mathcal{X}_-(H)$ be the stable $H$ invariant subspace and

$$\mathcal{X}_-(H) = \operatorname{Im} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

where $X_1, X_2 \in \mathbb{C}^{n\times n}$. If $X_1$ is nonsingular, then $X := X_2X_1^{-1}$ is uniquely determined by $H$, denoted by $X = Ric(H)$.

A key result of this chapter is the relationship between the spectral factorization of a transfer function and the solution of a corresponding ARE. Suppose $(A, B)$ is stabilizable and suppose either $A$ has no eigenvalues on $j\omega$-axis or $P$ is sign definite (i.e., $P \geq 0$ or $P \leq 0$) and $(P, A)$ has no unobservable modes on the $j\omega$-axis. Define

$$\Phi(s) = \begin{bmatrix} B^*(-sI - A^*)^{-1} & I \end{bmatrix} \begin{bmatrix} P & S \\ S^* & R \end{bmatrix} \begin{bmatrix} (sI - A)^{-1}B \\ I \end{bmatrix}.$$

Then

$$\Phi(j\omega) > 0 \ \text{for all} \ 0 \le \omega \le \infty$$

$\Longleftrightarrow \exists$ a stabilizing solution $X$ to

$$(A - BR^{-1}S^*)^*X + X(A - BR^{-1}S^*) - XBR^{-1}B^*X + P - SR^{-1}S^* = 0$$

$\Longleftrightarrow$ the Hamiltonian matrix

$$H = \begin{bmatrix} A - BR^{-1}S^* & -BR^{-1}B^* \\ -(P - SR^{-1}S^*) & -(A - BR^{-1}S^*)^* \end{bmatrix}$$

has no $j\omega$-axis eigenvalues.

Similarly,

$$\Phi(j\omega) \ge 0 \ \text{for all} \ 0 \le \omega \le \infty$$

$\Longleftrightarrow \exists$ a solution $X$ to

$$(A - BR^{-1}S^*)^*X + X(A - BR^{-1}S^*) - XBR^{-1}B^*X + P - SR^{-1}S^* = 0$$

such that $\sigma(A - BR^{-1}S^* - BR^{-1}B^*X) \subset \overline{\mathbb{C}}_-$.

Furthermore, there exists a $M \in \mathcal{R}_p$ such that

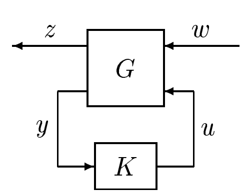
$$\Phi = M^\sim RM.$$

with

$$M = \left[\begin{array}{c|c} A & B \\ \hline -F & I \end{array}\right], \quad F = -R^{-1}(S^* + B^*X).$$

Chapter 14 treats the optimal control of linear time-invariant systems with quadratic performance criteria, i.e., LQR and $\mathcal{H}_2$ problems. We consider a dynamical system described by an LFT with

$$G(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & D_{12} \\ C_2 & D_{21} & 0 \end{array}\right].$$



Define

$$H_2 := \begin{bmatrix} A & 0 \\ -C_1^*C_1 & -A^* \end{bmatrix} - \begin{bmatrix} B_2 \\ -C_1^*D_{12} \end{bmatrix} \begin{bmatrix} D_{12}^*C_1 & B_2^* \end{bmatrix}$$

$$J_2 := \begin{bmatrix} A^* & 0 \\ -B_1B_1^* & -A \end{bmatrix} - \begin{bmatrix} C_2^* \\ -B_1D_{21}^* \end{bmatrix} \begin{bmatrix} D_{21}B_1^* & C_2 \end{bmatrix}$$

$$X_2 := Ric(H_2) \geq 0, \quad Y_2 := Ric(J_2) \geq 0$$

$$F_2 := -(B_2^*X_2 + D_{12}^*C_1), \quad L_2 := -(Y_2C_2^* + B_1D_{21}^*).$$

Then the $\mathcal{H}_2$ optimal controller, i.e. the controller that minimizes $\|T_{zw}\|_2$, is given by

$$K_{opt}(s) := \left[\begin{array}{c|c} A + B_2F_2 + L_2C_2 & -L_2 \\ \hline F_2 & 0 \end{array}\right].$$

Chapter 15 solves a max-min problem, i.e., a full information (or state feedback) $\mathcal{H}_\infty$ control problem, which is the key to the $\mathcal{H}_\infty$ theory considered in the next chapter. Consider a dynamical system

$$\begin{aligned} \dot{x} &= Ax + B_1w + B_2u \\ z &= C_1x + D_{12}u, \quad D_{12}^* \begin{bmatrix} C_1 & D_{12} \end{bmatrix} = \begin{bmatrix} 0 & I \end{bmatrix}. \end{aligned}$$

Then we show that $\sup_{w \in \mathcal{BL}_{2+}} \min_{u \in \mathcal{L}_{2+}} \|z\|_2 < \gamma$ if and only if $H_\infty \in dom(Ric)$ and $X_\infty = Ric(H_\infty) \geq 0$ where

$$H_\infty := \begin{bmatrix} A & \gamma^{-2}B_1B_1^* - B_2B_2^* \\ -C_1^*C_1 & -A^* \end{bmatrix}$$

Furthermore, $u = F_\infty x$ with $F_\infty := -B_2^*X_\infty$ is an optimal control.

Chapter 16 considers a simplified $\mathcal{H}_\infty$ control problem with the generalized plant $G(s)$ as given in Chapter 14. We show that there exists an admissible controller such that $\|T_{zw}\|_\infty < \gamma$ iff the following three conditions hold:

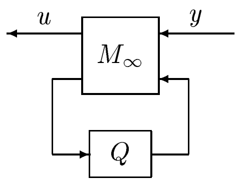
(i) $H_\infty \in$ dom(Ric) and $X_\infty := Ric(H_\infty) \geq 0$;

(ii) $J_\infty \in$ dom(Ric) and $Y_\infty := Ric(J_\infty) \geq 0$ where

$$J_\infty := \begin{bmatrix} A^* & \gamma^{-2}C_1^*C_1 - C_2^*C_2 \\ -B_1B_1^* & -A \end{bmatrix}.$$

(iii) $\rho(X_\infty Y_\infty) < \gamma^2$ .

Moreover, the set of all admissible controllers such that $\|T_{zw}\|_\infty < \gamma$ equals the set of all transfer matrices from $y$ to $u$ in




$$M_\infty(s) = \left[\begin{array}{c|cc} \hat{A}_\infty & -Z_\infty L_\infty & Z_\infty B_2 \\ \hline F_\infty & 0 & I \\ -C_2 & I & 0 \end{array}\right]$$

where $Q \in \mathcal{RH}_\infty$, $\|Q\|_\infty < \gamma$ and

$$\hat{A}_\infty := A + \gamma^{-2} B_1 B_1^* X_\infty + B_2 F_\infty + Z_\infty L_\infty C_2$$

$$F_\infty := -B_2^* X_\infty, \quad L_\infty := -Y_\infty C_2^*, \quad Z_\infty := (I - \gamma^{-2} Y_\infty X_\infty)^{-1}.$$

Chapter 17 considers again the standard $\mathcal{H}_\infty$ control problem but with some assumptions in the last chapter relaxed. We indicate how the assumptions can be relaxed to accommodate other more complicated problems such as singular control problems. We also consider the integral control in the $\mathcal{H}_2$ and $\mathcal{H}_\infty$ theory and show how the general $\mathcal{H}_\infty$ solution can be used to solve the $\mathcal{H}_\infty$ filtering problem. The conventional Youla parameterization approach to the $\mathcal{H}_2$ and $\mathcal{H}_\infty$ problems is also outlined. Finally, the general state feedback $\mathcal{H}_\infty$ control problem and its relations with full information control and differential game problems are discussed.

Chapter 18 first solves a gap metric minimization problem. Let $P = \tilde{M}^{-1}\tilde{N}$ be a normalized left coprime factorization. Then we show that

$$\inf_{K \ stabilizing} \left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \begin{bmatrix} I & P \end{bmatrix} \right\|_\infty$$

$$= \inf_{K \ stabilizing} \left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \right\|_\infty = \left( \sqrt{1 - \left\| \begin{bmatrix} \tilde{N} & \tilde{M} \end{bmatrix} \right\|_H^2} \right)^{-1}.$$

This implies that there is a robustly stabilizing controller for

$$P_\Delta = (\tilde{M} + \tilde{\Delta}_M)^{-1}(\tilde{N} + \tilde{\Delta}_N)$$

with

$$\left\| \begin{bmatrix} \tilde{\Delta}_N & \tilde{\Delta}_M \end{bmatrix} \right\|_\infty < \epsilon$$

if and only if

$$\epsilon \le \sqrt{1 - \left\| \begin{bmatrix} \tilde{N} & \tilde{M} \end{bmatrix} \right\|_H^2}.$$

Using this stabilization result, a loop shaping design technique is proposed. The proposed technique uses only the basic concept of loop shaping methods and then a robust stabilization controller for the normalized coprime factor perturbed system is used to construct the final controller.

Chapter 19 considers the design of reduced order controllers by means of controller reduction. Special attention is paid to the controller reduction methods that preserve the closed-loop stability and performance. In particular, two $\mathcal{H}_\infty$ performance preserving reduction methods are proposed:

a) Let $K_0$ be a stabilizing controller such that $\|\mathcal{F}_\ell(G,K_0)\|_\infty < \gamma$. Then $\hat{K}$ is also a stabilizing controller such that $\left\|\mathcal{F}_\ell(G,\hat{K})\right\|_\infty < \gamma$ if

$$\left\|W_2^{-1}(\hat{K}-K_0)W_1^{-1}\right\|_\infty < 1$$

where $W_1$ and $W_2$ are some stable, minimum phase and invertible transfer matrices.

b) Let $K_0 = \Theta_{12}\Theta_{22}^{-1}$ be a central $\mathcal{H}_\infty$ controller such that $\|\mathcal{F}_\ell(G,K_0)\|_\infty < \gamma$ and let $\hat{U},\hat{V} \in \mathcal{RH}_\infty$ be such that

$$\left\|\begin{bmatrix} \gamma^{-1}I & 0 \\ 0 & I \end{bmatrix}\Theta^{-1}\left(\begin{bmatrix} \Theta_{12} \\ \Theta_{22} \end{bmatrix} - \begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix}\right)\right\|_\infty < 1/\sqrt{2}.$$

Then $\hat{K} = \hat{U}\hat{V}^{-1}$ is also a stabilizing controller such that $\|\mathcal{F}_\ell(G,\hat{K})\|_\infty < \gamma$.

Thus the controller reduction problem is converted to weighted model reduction problems for which some numerical methods are suggested.

Chapter 20 briefly introduces the Lagrange multiplier method for the design of fixed order controllers.

Chapter 21 discusses discrete time Riccati equations and some of their applications in discrete time control. Finally, the discrete time balanced model reduction is considered.